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RenovAI: a generative artificial intelligence platform for residential renovation visualisation - a computational solution and business plan for the Brazilian market

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Final Course Project submitted to the
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“Recognising the need is the primary condition for design.” (Charles Eames)

RESUMO

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A RenovAI é uma plataforma digital brasileira de inteligência artificial generativa para o planejamento de reformas residenciais, voltada ao público geral e não a arquitetos profissionais. A partir de uma descrição ou fotografia de um cômodo, a plataforma gera uma imagem fotorrealista do ambiente reformado e uma lista de móveis sugeridos, em poucos minutos e sem custo. O trabalho apresenta o desenvolvimento da solução computacional, baseada em tecnologias web modernas e em serviços de inteligência artificial generativa em nuvem, e o respectivo plano de negócios. O problema endereçado é a incerteza antes do gasto: 82% das moradias brasileiras são feitas sem profissional habilitado (CAU/Datafolha, 2022) e o projeto de arquitetura custa de R\$ 55 a R\$ 140 por metro quadrado. O modelo de negócios baseia-se em comissões sobre vendas, com planos pagos opcionais. A validação técnica foi conduzida com usuários reais em produção. Conclui-se que a solução é tecnicamente robusta e que o modelo é promissor, condicionado à validação das hipóteses de conversão com dados próprios.

Palavras-chave: inteligência artificial generativa; design de interiores; reforma residencial; plano de negócios; plataforma SaaS.

ABSTRACT

WASSERSTEIN, Bruno. RenovAI: a generative artificial intelligence platform for residential renovation visualisation. 2026. 19 p. Final Course Project (Bachelor) - Computer Science, Institute of Technology and Leadership (INTELI), São Paulo, 2026.

RenovAI is a Brazilian generative artificial intelligence platform for residential renovation planning, aimed at the general public rather than professional architects. From a description or photograph of a room, the platform generates a photorealistic image of the renovated space and a list of suggested furniture, in minutes and at no cost. This work presents the development of the computational solution, based on modern web technologies and cloud generative-AI services, and the corresponding business plan. The problem addressed is uncertainty before spending: 82% of Brazilian homes are built without a qualified professional (CAU/Datafolha, 2022), and an architecture project costs from R\$ 55 to R\$ 140 per square metre. The business model is based on commissions on sales, with optional paid plans. Technical validation was carried out with real users in production. The solution is found to be technically robust and the model promising, contingent on validating the conversion hypotheses with proprietary data.

Keywords: generative artificial intelligence; interior design; residential renovation; business plan; SaaS platform.

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LIST OF ABBREVIATIONS AND ACRONYMS

AI	Artificial Intelligence
MVP	Minimum Viable Product
TAM/SAM/SOM	Total / Serviceable Available / Serviceable Obtainable Market
BMC	Business Model Canvas
CDN	Content Delivery Network
LGPD	Brazilian General Data Protection Law
SWOT	Strengths, Weaknesses, Opportunities and Threats

SUMMARY

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1 Introduction

Deciding to renovate or furnish a home is, for most Brazilians, a high-stakes financial decision made without technical vocabulary and without the budget to hire an architect. RenovAI is a generative artificial intelligence platform that lets anyone visualise a renovated room before spending a single cent, addressing a structural gap in access to interior-design services in Brazil.

Context and Motivation.

Architecture and interior-design services in Brazil remain elitist. A residential architecture project costs on average between R\$ 55 per square metre (GetNinjas, 2024) and R\$ 140 per square metre in renovations (Habitissimo, 2024) - equivalent to thousands of reais for a single room. As a result, only 18% of Brazilians hire an architect or engineer for their own construction work, meaning 82% of homes are built without a qualified professional (CAU/Datafolha, 2022). Generative AI created, for the first time, the technical possibility of democratising this access in a scalable way, at a marginal cost of a few cents per visualisation.

Problem Definition and Value Proposition.

The core problem is not aesthetics but uncertainty before spending: the fear of getting a renovation wrong. RenovAI's value proposition is to deliver visual certainty before the purchase decision - a photorealistic image of the renovated space plus a curated, priced furniture list - for free, on a mobile phone, in minutes, requiring no technical knowledge from the user.

Scope and positioning.

RenovAI operates in the space between two extremes: doing everything alone, without the technical repertoire to visualise the result, and hiring a complete professional project, with its cost, time and access barriers. The platform offers light, exploratory and accessible planning between these poles.

Table 1 - Positioning between two extremes

Extreme	Perceived limitation	Space occupied by RenovAI
Doing everything alone	Lack of technical repertoire, insecurity, risk of buying wrong, difficulty visualising the result	Visual support, organisation of preferences, suggestions and simulations
Hiring a complete professional project	Cost, time, bureaucracy, access barrier and a scope larger than desired	Light, exploratory and accessible planning

Source: prepared by the author (2026).

RenovAI is not designed to replace architects, interior designers, engineers or professionals qualified for technical interventions. Its scope is light renovation, decoration, furniture, painting, spatial organisation and preliminary visualisation. Structural interventions, complex electrical or hydraulic work, demolitions, facade changes or any activity requiring formal technical responsibility are explicitly outside the product's safe scope, and the interface communicates this limit to the user.

The decisions the product helps with.

Many people want to improve a room but do not know how to turn loose references into a practical decision. In practice, the user typically faces questions such as:

- Does this piece of furniture suit the space?
- Will this colour work?
- Will the room look cramped?
- How do I compare different styles?
- Is it worth spending on a particular item?
- How do I explain to someone else what I want?
- How do I visualise it before buying?

RenovAI answers these through a guided experience: the user provides data about the room, the system understands the goal of the renovation, the AI generates a visual proposal, the platform organises the suggestions in a more actionable form, and the user can review, compare versions and evolve the project.

SIMPLIFIED USER FLOW

```
The user provides data about the room
-> the system understands the goal
-> the AI generates a visual proposal
-> the platform organises actionable suggestions
-> the user reviews, compares versions and evolves the project
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Target audience.

The main audience is people who are not experts in architecture, design or construction, especially users who want to renovate, decorate or reorganise rooms without starting a full project with a high degree of technical complexity. Several profiles were considered during the project.

Table 2 - User profiles considered

Profile	Need	Relationship with RenovAI
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Ordinary user renovating a room	Visualise before spending	Primary audience
Person without a design repertoire	Receive simple guidance	Primary audience
User with Pinterest/Instagram references	Turn inspiration into an applicable proposal	Primary audience
Couples or families deciding together	Compare preferences and reduce conflict	Explored idea, partly conceptual
Condominium residents	Standardisation and planning of light improvements	Explored vertical
Retailers / partners	Connect a catalogue to generated rooms	Planned/partly prototyped vertical
Architecture/design professionals	Use as an ideation aid, not a replacement	Secondary or future audience

Source: prepared by the author (2026).

Methodology overview.

The work followed an applied, iterative methodology combining problem exploration, technology validation, the construction of a functional MVP, the expansion of modules and academic consolidation, supported throughout by AI-assisted engineering. The full development process, tools and phases are detailed in Section 2.4.

Objectives of the Work.

The general objective is to develop and validate a computational solution for AI-assisted residential renovation and to elaborate a business plan for its introduction into the market. The specific objectives are:

- build a functional MVP that generates a photorealistic visualisation and a furniture list from a user briefing;
- validate the product technically with real users in production;
- define a revenue model for the solution;
- elaborate a complete business plan, including market sizing, competitive analysis and financial feasibility.

Justification and Contributions.

The addressable market - annual household spending on home goods in Brazil - is estimated at R\$ 92 billion. Beyond the economic relevance, the solution has documented social-impact potential aligned with the United Nations Sustainable Development Goals 1, 10 and 11, by lowering the cost of design decisions for low-income households that renovate the home they already have. The technical contribution is a production-grade, fully localised generative pipeline for the Brazilian market (prices in reais, Brazilian brands, native Portuguese).

Structure of the Work.

Chapter 1 introduces the problem and objectives. Chapter 2 develops the solution: it defines the market assumptions and hypotheses (2.1), sizes and analyses the market (2.2), examines the competition (2.3), describes the technological solution (2.4), presents the business plan (2.5) and reports the market validation and results (2.6). Chapter 3 concludes the work. The References, Appendices and Annexes close the document.

Table 3 - Project overview

Aspect	Description
Product stage	MVP in production at renovai.ia.br
Target audience	General public planning a residential renovation
Business model	Commission on sales, with optional paid plans
Reference market	Home-goods spending by Brazilian urban households
Operating structure	100% digital, cloud-native, lean fixed cost
Funding	Bootstrapped by the founder; non-dilutive public grants under consideration

Source: prepared by the author (2026).

2 Solution Development

This chapter develops the RenovAI solution end to end. It begins with the strategic assumptions and hypotheses that guided the product (2.1), quantifies the market opportunity (2.2) and the competitive landscape (2.3), details the technological solution actually built and running in production (2.4), consolidates the business plan (2.5), and reports the validation carried out with real users (2.6).

2.1 Definition of Market Assumptions and Hypotheses

The product was designed around three explicit, falsifiable hypotheses, each to be confirmed or refuted with proprietary data collected from real usage.

Problem Hypothesis.

Most people planning a renovation face a real pain point - the fear of buying wrong - and do not hire a professional, given the cost. The assumption is that this public will adopt a free tool that delivers visual certainty before they spend.

Solution Hypothesis.

The assumption is that a pure generative-AI solution - instant, mobile-first, with a furniture list linked to real stores - is a better way to solve the problem than the existing alternatives (human-in-the-loop services that cost from R\$ 149 and take days, or web-only AI tools without a shoppable list).

Value Hypothesis.

The assumption is that the revenue model is acceptable to users: the visualisation is free, monetised indirectly through commissions on furniture the user would buy anyway, with optional paid plans for heavier users.

2.2 Market Sizing and Analysis

Market size was estimated bottom-up, with each factor traceable to a published primary source.

Market Size (TAM, SAM, SOM).

The Total Addressable Market is annual spending by Brazilian urban households on home goods. The Serviceable Available Market restricts it to households with purchasing power and service consideration. The Serviceable Obtainable Market is RenovAl's conservative three-year potential, an internal assumption to be validated.

Table 4 - Market sizing (TAM, SAM, SOM)

Layer	Value	Methodology / source
TAM	R\$ 92 billion/year	67.7 million urban households (FJP, 2023) × R\$ 1,362 average home-goods spend (ABCasa/IEMI, 2024)
SAM	R\$ 30.8 billion/year	TAM × 45.7% (classes A-C1, ABEP, 2024) × 73% (would consider an architect, CAU/Datafolha, 2022)
SOM (year 3)	R\$ 8.5 million/year	0.5% penetration of the SAM × hybrid ARPU of R\$ 75/year - internal assumption to validate

Source: prepared by the author (2026), based on the cited sources.

Market context.

The Brazilian context reinforces the opportunity. The home-goods sector moved approximately R\$ 102 billion in 2024 (ABCasa/IEMI, 2024), and a large share of households live in homes with some form of inadequacy: the Fundação João Pinheiro estimates millions of urban households with qualitative housing inadequacy and a quantitative housing deficit (FJP, 2024). Combined with the fact that 82% of homes are built without a qualified professional (CAU/Datafolha, 2022), this points to a large base of people who renovate the home they already have, largely on their own - precisely the public RenovAl serves.

Table 5 - Public context indicators

Indicator	Value	Source
Home-goods sector (2024)	~R\$ 102 billion	ABCasa/IEMI, 2024
Homes built without a qualified professional	82%	CAU/Datafolha, 2022

Average residential architecture cost	R\$ 55 to R\$ 140 per square metre	GetNinjas, 2024; Habitissimo, 2024
Urban households with housing inadequacy/deficit	Millions of households	FJP, 2024

Source: prepared by the author (2026), based on the cited sources.

Customer Segmentation and Personas.

RenovAl serves several customer segments, from mass-market renovators to higher-income owners of compact properties, in addition to local artisans as future marketplace suppliers.

Table 6 - Customer personas

Persona	Profile	Job to be done
A - Decided Renovator	Adult, middle class, renovating a room	Certainty before spending
B - Urban Dweller	Young adult, first apartment	Translate inspiration into a decision
C - Compact High-End Owner	Young high-income, compact flat	Quality result fast, no waiting
D - Local Artisan	Artisan, future marketplace supplier	Reach buyers with a defined project

Source: prepared by the author (2026).

Persona detail.

Persona A, the decided renovator, is the core audience: an adult, often middle class, who is going to renovate or furnish a room and whose main pain is the fear of buying wrong, having never hired an architect; they need certainty before spending and decide carefully, with high price sensitivity.

Persona B, the urban dweller, is a younger adult, frequently in a first apartment, who has a clear aesthetic vision but struggles to articulate it for execution; they are more experimental and often share the result on social media, seeking to translate diffuse inspiration into a concrete decision.

Persona C, the owner of a compact high-end property, is a young, higher-income user living in a small flat in a valued neighbourhood; sensitive to time rather than price, they want a quality result quickly, without waiting weeks for a project.

Persona D, the local artisan, is not a design buyer but the supplier of the future marketplace: a carpenter, artisan or product designer who lacks a qualified sales channel and would benefit from reaching buyers who already have a defined visual project.

2.3 Competitive Analysis and Differentials

A mapping conducted in May 2026 identified 27 relevant players. The Brazilian AI-assisted interior-design market is more crowded than a first read suggests, but the segments differ.

Direct and Indirect Competitors.

Direct Brazilian competitors include both AI-only tools and human-assisted services (such as Arquiteto de Bolso, DecoraGPT, IAdecora and Archie). International references that frame the category include Wayfair (with its Decorify/Muse AI tools, but absent from Brazil), Houzz, RoomGPT and Havenly, while the closure of the pure-human Mody illustrates that human-in-the-loop models struggle to scale.

Table 7 - Competitive comparison (overview)

Criterion	RenovAI	AI-only tools	Human-assisted services
Price	Free (MVP)	Free / paid	From R\$ 149
Delivery	AI, under a minute	AI, minutes	Human, days
Localisation (PT-BR, BRL)	Yes	Varies	Yes
Mobile-first experience	Yes	Mostly web	Mostly web

Source: prepared by the author (2026).

International benchmarks.

Five international references frame the possible space for the category and help calibrate expectations.

Table 8 - International benchmarks

Reference	Country	Model	Lesson for RenovAI
Wayfair (Decorify/Muse)	USA	Retailer with integrated generative AI; not present in Brazil	Validates the 'AI visualisation + shoppable products' category; the Brazilian window is open
Houzz	USA	Marketplace, tools and a professional community	Shows the possible ceiling when community and catalogue become the moat
RoomGPT	USA (indie)	Pure-AI, web-only, freemium	Proves demand for pure-AI generation, with a positive exit
Havenly	USA	AI plus human plus a marketplace	The highest-monetisation model; combines speed with curation
Mody	USA (closed 2022)	Pure-human 3D rendering	Shows that human-in-the-loop models struggle to scale

Source: prepared by the author (2026).

RenovAI positions itself consciously between a pure-AI tool (speed) and a curated marketplace, without the marginal cost of human designers and adapted to the

Brazilian market - a combination not occupied by the international references, which do not serve Brazil natively.

Competitive Advantage and Differentiating Factor.

Generative AI itself is an increasingly commoditised resource; RenovAI differentiates through deep localisation for the Brazilian market (prices in reais, local brands, native Portuguese), a mobile-first experience and ease of use for non-experts, supported by a low operating cost.

2.4 Technological Solution

RenovAI is a fully digital, mobile-first platform in production at renovai.ia.br. This section documents how it was built: the development methodology and the journey across roughly nine months, the requirements and use cases, the system architecture and data model, the development of each module, and the testing and technical evaluation - including the testing that was planned but not implemented.

Design principles.

A set of principles guided the technical and product decisions:

- guide the lay user, reducing the need for free prompting;
- keep the core flow simple even when the backend is complex;
- preserve the original room and avoid structural promises;
- store versions so the user can compare alternatives;
- abstract AI providers and allow fallback;
- treat the catalogue as a data system, not just text;
- protect personal data and communicate the product's limits.

Development methodology and process

Because the project was led mainly by a single founder-developer in a context of high technical uncertainty, the formal methodology was adapted to a reality of a lean team, rapid prototyping and constant technical validation. In practice the process combined elements of Scrum, informal Kanban and exploratory, prototype-driven development, with intensive use of AI assistants as engineering tools.

Table 9 - Methods used in the project

Method	How it appeared in the project
Academic Scrum	Organisation by sprints, incremental deliveries and documentation by phase
Informal Kanban	A continuous list of tasks, bugs, improvements and hypotheses

Rapid prototyping	Screens, flows and agents built to test feasibility
AI-assisted development	Intensive use of ChatGPT, Claude Code, Cursor and generative models
Exploratory validation	Manual testing, real usage and collection of user perception
Evolutionary architecture	Gradual change of backend, database, frontend and modules as the product matured

Source: prepared by the author (2026).

The development did not follow an industrial process with perfect documentation from the start; it was marked by cycles of discovery, attempt, discard, rewrite and consolidation, which is coherent with a technological-innovation project, especially in generative AI, where technical uncertainty is high. The work can be organised into five phases.

Table 10 - Phases of the project

Phase	Focus	Result
1	Problem exploration	Definition of the pain, the audience and the initial proposition
2	Technical validation	Tests with language models, image generation, catalogue and agents
3	Functional MVP	Project flow, visual generation, history and website
4	Module expansion	AI Designer, personas, retailer portal, condominiums, 3D
5	Academic consolidation	Documentation, auditing and technical/narrative refinement

Source: prepared by the author (2026).

Phase 1 - Problem exploration.

The focus was understanding the pain of an ordinary user when renovating or decorating. A central thesis emerged: the greatest pain is not a lack of inspiration, but the lack of translation between inspiration and decision. Key learnings were that the user wants confidence rather than just a pretty image; that a light renovation involves emotional, aesthetic, financial and practical decisions; that the AI must be guided by structured questions rather than free prompts; and that the solution must not look like a technical tool for professionals.

Phase 2 - Technology validation.

This phase investigated whether textual and visual data about a room could be turned into useful proposals. It explored language models for briefing interpretation, image-generation models, specialised agents, product recommendation, catalogue search, layout generation, simplified 3D rendering, structured JSON outputs, a relational database and embeddings for semantic similarity. An initial experimental architecture (FastAPI, a standalone PostgreSQL database and Docker, with agent services) was built to validate endpoints and recommendation; a known technical issue was aligning embedding dimensionality and the consistency between the models used

for indexing and search. The main learning was that isolated image generation does not solve the problem - the real challenge is connecting image, briefing, budget, style and real products, while preserving versions so the user can compare alternatives.

Phase 3 - Functional MVP.

The experiments were turned into a usable web platform with a main flow of project creation, result visualisation and history. It was in this phase that the architecture migrated from the initial experimental backend to the current serverless stack. Important decisions were to focus on light renovation (reducing technical and legal risk); to be mobile-first (the user photographs the room with a phone); to use native Brazilian Portuguese; to store versions (renovations involve comparing alternatives); to avoid any structural promise; and to separate inspiration from execution. Learnings were that the first generated image has a strong emotional impact but must be explainable, that the history is essential because a project is not decided in a single session, and that a good interface reduces the user's dependence on prompt engineering.

Table 11 - Key decisions of the MVP phase

Decision	Justification
Focus on light renovation	Reduces technical and legal risk
Mobile-first	The user photographs the room with a phone
Native Brazilian Portuguese	A Brazilian product with accessible language
Store versions	Renovations involve comparing alternatives
Avoid any structural promise	Safety and technical responsibility
Separate inspiration from execution	The product should help decide, not only generate a pretty image

Source: prepared by the author (2026).

Phase 4 - Module and vertical expansion.

After the MVP, the project explored more ambitious modules: a conversational AI Designer, architect personas, a debate between styles, a retailer portal, a condominium vertical, a Layout Studio 3D, AR/VR experiments, a moodboard and analytics. This phase was exploratory: some ideas became functional interfaces or routes, others remained concepts. The main learning was that a multi-agent approach is useful for ideation but can add excessive complexity for a lay user, and that the product must avoid becoming a scattered collection of demos.

Phase 5 - Academic consolidation.

The current phase turns the project into academic and technical material: documentation of the architecture, a product audit, organisation of the modules, an

analysis of testing, definition of future work and preparation of this report. The challenge is to document RenovAI honestly, without overstating what is still a prototype and without revealing sensitive business detail.

Tools and technologies.

Beyond the code stack (detailed in the architecture subsection), the project used the following tools.

Table 12 - Development and product tools

Tool	Use
ChatGPT	Ideation, documentation, prompts, technical and product decisions
Claude Code	Implementation, refactoring and code reading/auditing
Cursor	AI-assisted development
Git / GitHub	Version control and code organisation
Vercel dashboard	Deployment and build monitoring
Supabase dashboard	Database, authentication, tables and policies
Spreadsheets / CSV	Catalogue structuring and research

Source: prepared by the author (2026).

How AI assisted the development.

Artificial intelligence was used not only as a product feature but also as a development tool. It assisted in creating React components, analysing build errors, writing endpoints, reviewing architecture, creating seed scripts, debugging database problems, generating internal prompts, organising types and entities, writing documentation, comparing technical alternatives and planning tests.

Table 13 - Examples of AI-assisted engineering

Situation	How AI helped
Initial backend structuring	Suggested an organisation with experimental services, a database and agents
Product catalogue	Helped create and standardise CSV fields
Deployment problems	Diagnosed scripts, runtime versions and build configuration
Image prompts	Refined instructions to produce more coherent rooms
Report and documentation	Structured the narrative and the technical documentation
Product audit	Analysed the flow, UX, inconsistencies and maturity

Source: prepared by the author (2026).

AI was also used to simulate the perspectives of different users and specialists (a lay user, an architect, an interior designer, a UX auditor, an AI engineer, a retailer and a sceptical user), which allowed RenovAI to be evaluated from multiple angles without a full multidisciplinary team at every step.

Process limitations.

Table 14 - Limitations of the process

Limitation	Impact
Lean team	Lower capacity for continuous testing, documentation and QA
High dependence on generative AI	Output variability and difficulty of control
Product spread across many fronts	Risk of dispersion between modules
Little test automation	Higher risk of regressions
Architecture changing over time	Harder to trace older decisions
Dependence on external APIs	Risk of cost, availability and model changes
Limited broad validation	Hypotheses still require more testing with users

Source: prepared by the author (2026).

Requirements and specifications

The system requirements were derived from the product vision and refined throughout development. The tables below list the functional and non-functional requirements with their current status (concluded, partial, prototype, planned or not implemented).

Table 15 - Functional requirements

ID	Requirement	Status
RF01	User sign-up and login	Concluded/partial
RF02	Create a renovation or decoration project	Concluded/partial
RF03	Select the room type	Concluded/partial
RF04	Collect style preferences (style, colours, atmosphere)	Concluded/partial
RF05	Upload a photo of the room	Partial/concluded
RF06	Describe the room textually, without a photo	Partial/concluded
RF07	Generate a visual proposal with AI	Concluded/partial
RF08	Save the generated result and project data	Concluded/partial
RF09	Access a history of previous projects	Concluded/partial
RF10	Organise projects by property	Partial
RF11	Organise projects by room	Partial
RF12	Create and view alternative versions of a project	Partial
RF13	Recommend furniture compatible with the room and style	Partial/prototype
RF14	Display the recommended product list with basic information	Partial
RF15	Link to external products when available	Partial/to confirm
RF16	Export a project summary/budget as a PDF	Partial/to confirm
RF17	Chat with an AI assistant	Prototype/partial
RF18	Simulate an evaluation by multiple personas	Prototype
RF19	Retailer portal for product registration/management	Planned/draft
RF20	3D spatial visualisation of the room	Prototype
RF21	Generate or present a style moodboard	Partial/planned
RF22	Record key usage events (analytics)	Partial/to confirm
RF23	Inform that the product does not replace a professional project	Partial/concluded
RF24	Handle generation errors and allow a retry	Partial/to confirm
RF25	Allow the user to delete projects	To confirm

Source: prepared by the author (2026).

Table 16 - Non-functional requirements

ID	Requirement	Status
RNF01	Usability for lay users, without architectural jargon	Partial/concluded
RNF02	Responsiveness on phones and desktops	Partial/concluded
RNF03	Clear loading states during long operations	Partial
RNF04	Availability on reliable hosting for public access	Partial/concluded
RNF05	Security: users cannot access other users' data	Partial/concluded
RNF06	Privacy: careful handling of home images and data	Partial/to confirm
RNF07	Scalability without a full rewrite	Partial
RNF08	Observability of failures and errors	Partial/to confirm
RNF09	Maintainability through modular, reusable code	Partial
RNF10	Reliability of AI outputs (consistent and aligned with the briefing)	Partial
RNF11	Accessibility	Planned/not formally validated
RNF12	Cross-browser compatibility	Planned/partial; formal tests not confirmed

Source: prepared by the author (2026).

Use cases

The main use cases describe how different actors interact with the platform.

Table 17 - Main use cases

Use case	Actor	Summary of the flow	Status
Create a bedroom project	Ordinary user	Log in, choose the room, enter preferences, send photo/description, generate, save and revisit in the history	Concluded/partial
Compare versions of a living room	Undecided user	Open an existing project, request a new version in another style, compare the two and decide a direction	Partial
Receive furniture suggestions	User who wants to buy	Generate the room; the system identifies the needed categories, recommends items and the user reviews the list and external links	Partial/prototype
Organise projects by property	User with several rooms	Create a property, add rooms, generate versions per room and browse the history by property	Partial
Talk to the AI Designer	User unsure what they want	Open the AI Designer, describe the problem; the agent asks questions and the system summarises preferences and generates a proposal	Prototype/partial
Evaluation by personas	User wanting a critical opinion	Submit a project; personas evaluate and the system summarises conflicts and consensus into a final recommendation	Prototype
Retailer registers products	Retailer	Access the portal, register the store, submit a catalogue; products become available for recommendation	Planned/draft
Manager views a common-area improvement	Building manager	Inform a common area; the platform generates a visual proposal used for internal discussion	Concept/partial

Source: prepared by the author (2026).

The main use cases are detailed below, each with its actor, the steps of the flow and the current status.

Use case 1 - Create a bedroom project.

Actor: an ordinary user.

- The user accesses the platform.
- Logs in or starts a project.
- Chooses the room type.
- Enters preferences.
- Sends a photo or a description.
- Requests the generation.
- Receives the image.
- Saves the project.
- Revisits it later in the history.

Status: concluded/partial.

Use case 2 - Compare versions of a living room.

Actor: an undecided user.

- Opens an existing project.
- Requests a new version.
- Chooses another style.
- Compares the previous and the new image.
- Decides on a direction to follow.

Status: partial.

Use case 3 - Receive furniture suggestions.

Actor: a user who wants to buy items.

- Generates the room.
- The system identifies the needed categories.
- The system recommends items.
- The user reviews the list.
- The user accesses external links.

Status: partial/prototype.

Use case 4 - Organise projects by property.

Actor: a user with more than one room or property.

- Creates a property.
- Adds rooms.
- Generates versions per room.
- Consults the history by property.

Status: partial.

Use case 5 - Talk to the AI Designer.

Actor: a user unsure of what they want.

- Opens the AI Designer.
- Describes the problem.
- The agent asks questions.
- The system summarises the preferences.
- The user generates a proposal.

Status: prototype/partial.

Use case 6 - Evaluation by personas.

Actor: a user who wants a critical opinion.

- Submits a project.
- The personas evaluate it.
- The system summarises conflicts and consensus.
- The user receives a final recommendation.

Status: prototype.

Use case 7 - Retailer registers products.

Actor: a retailer.

- Accesses the portal.
- Registers the store.
- Submits a catalogue.
- The products become available for recommendation.

Status: planned/draft.

Use case 8 - Manager views a common-area improvement.

Actor: a building manager.

- Informs the common area.
- The platform generates a visual proposal.
- The result is used for internal discussion.
- Next steps are decided outside the platform.

Status: concept/partial.

System architecture

RenovAI is a mobile-first web application supported by services for authentication, database, storage, internal APIs and external generative-AI services. A simplified view of the layers is shown below.

ARCHITECTURE (logical layers)

Web / mobile frontend

- > Authentication and session layer
- > Application layer / internal APIs
- > Relational database
- > AI services (image generation, briefing interpretation, recommendation)
- > Storage of results
- > Review, history and export interface

The architecture evolved over the project. An initial experimental backend (FastAPI, a standalone PostgreSQL database and Docker, with agent services) was used in the technology-validation phase. The production architecture then migrated to a serverless stack, which lowered operational complexity and cost and improved deployment speed. The current stack is summarised below.

Table 18 - Current technology stack

Layer	Technology	Status
Frontend	React 18 + TypeScript (strict) + Vite	Concluded/in production
Styling / UI	Tailwind CSS + shadcn/ui + Radix UI	Concluded/in production
State / forms	Zustand; React Hook Form + Zod	Concluded/partial
3D	Three.js + React Three Fiber	Prototype
Backend	Vercel Edge / serverless functions	Concluded/in production
Database / Auth / Storage	Supabase (PostgreSQL, Auth, Storage)	Concluded/in production
Rate limiting	Vercel KV (Upstash Redis)	Concluded/partial
Deployment	Vercel	Concluded/in production
Image generation	Google Gemini (premium model), with Imagen 4 and Flux as fallback	Concluded/partial
Text / vision AI	Google Gemini (briefing, vision, agents)	Concluded/partial
Earlier experimental backend	FastAPI + PostgreSQL + Docker	Superseded (v1)

Source: prepared by the author (2026).

The main technology choices were guided by the need for a lean, low-cost and fast-to-iterate stack suitable for a solo founder.

Table 19 - Technology choices and rationale

Choice	Rationale
React + Vite + TypeScript	A fast, strongly typed frontend with quick builds and a large ecosystem
Serverless (Vercel Edge)	No servers to manage, automatic scaling and a low fixed cost
Supabase	Database, authentication and storage with row-level security in a single managed service
Managed Redis	Rate limiting without dedicated infrastructure
Multi-model AI with fallback	Availability and resilience against a single provider's failure or price change
Mobile-first	The user photographs the room with a phone
Native Brazilian Portuguese	A product localised for the Brazilian market

Source: prepared by the author (2026).

Application layers.

The presentation layer concentrates the landing page, the project-creation flow, the result screen, the logged-in area, the project history, the property/room/version pages, the AI Designer and the experimental panels. Its responsibilities are to guide the lay user, reduce the need for free prompting, display visual results, organise project data and communicate the product's limits. The application layer coordinates the business rules and AI calls: creating a project, updating the briefing, saving the original image, calling the language and image models, storing the result, creating a version, fetching the history, generating a budget and recommending furniture. The data layer stores users, properties, rooms, projects, versions, images, recommendations, products, chat messages, analytics events and generation logs.

The AI layer should be understood not as a single prompt but as a set of specialised steps. A simplified logical pipeline is shown below; the specific prompt-engineering techniques are a competitive differentiator and are intentionally kept generic in this public document.

```
AI PIPELINE (logical)

User input
-> normalisation
-> intent interpretation
-> constraint extraction
-> structured briefing composition
-> internal prompt generation
-> image-model call (with fallback)
-> post-processing / storage
-> item recommendation
-> explanation of the result
```

Data model

The data model reflects the real structure of a residential renovation. The main entities and their status are listed below.

Table 20 - Main entities

Entity	Description	Status
User	An authenticated user	Concluded/partial
Property	A registered home	Partial/concluded
Room	A room within a property	Partial/concluded
Project	A renovation or decoration project for a room	Concluded/partial
Version	A variant of a proposal	Partial/concluded
GeneratedImage	An AI-generated image	Concluded/partial
OriginalImage	A photo uploaded by the user	Partial
Briefing	User answers and preferences	Concluded/partial
Product	A catalogue item	Partial
Recommendation	A product recommended for a project	Partial/prototype
ChatSession / ChatMessage	A conversation with the AI	Partial
AgentPersona	An architect/designer persona	Prototype
Retailer	A partner store	Planned/draft
AnalyticsEvent	A usage event	Partial/to confirm

Source: prepared by the author (2026).

The relationships follow the Property to Room to Project to Version hierarchy, with images and recommendations attached to each version.

RELATIONSHIPS

```
User 1 -- N Property
Property 1 -- N Room
Room 1 -- N Project
Project 1 -- N Version
Version 1 -- N GeneratedImage
Version 1 -- N Recommendation
Recommendation N -- 1 Product
Project 1 -- N ChatSession
ChatSession 1 -- N ChatMessage
```

This hierarchy was an important product decision: a real renovation is not a single isolated image. A user may own more than one property, with several rooms per property, several proposals per room, successive revisions and versions for different budgets. The hierarchical structure helps the platform be perceived as a planning tool rather than merely an image generator. Access to properties and their children is mediated by a security-definer function that avoids policy recursion, and the data was backfilled idempotently so that legacy projects without a property were automatically linked to a default one.

Data dictionary and screens

To complement the data model, this subsection documents the main intake and product fields and the principal screens of the application. The intake (anamnesis) collects the data that guides the generation; the main fields and their purpose are listed below.

Table 21 - Intake fields

Field	Purpose	Status
Room type	Contextualise the layout and furniture	Concluded/partial
Dimensions	Improve the sense of scale	Partial
Ceiling height	Adjust spatial perception	Planned/partial
Doors and windows	Avoid unfeasible suggestions	Partial
Main use	Direct the room's function	Concluded/partial
Desired style	Guide the aesthetics	Concluded
Colour palette	Control the appearance	Partial
Mandatory items	Preserve the user's needs	Partial
Restrictions	Avoid incompatible solutions	Partial
Generic budget	Modulate the recommendations	Partial
Pets / allergies	Adjust materials and choices	Planned/partial
Room photo	Visual basis	Partial/concluded

Source: prepared by the author (2026).

The product catalogue uses standardised fields so that recommendations can be filtered, displayed and explained.

Table 22 - Product fields

Field	Description
Product name	Commercial name
Main category	For example sofa, bed or table
Subcategory	For example reclining, double or office
Store	Origin of the product
Current price	Value at the time of collection
Promotional price	When applicable
Dimensions	Item measurements
Predominant material	Wood, MDF, metal, fabric and so on
Colour / finish	Aesthetic information
Short description	Item summary
Link	Product URL
Associated style	To confirm
Compatible rooms	To confirm

Source: prepared by the author (2026).

The application is organised into a set of screens and routes, summarised below.

Table 23 - Main screens and routes

Area	Purpose
Landing page	Communicates the proposition to visitors
Sign-up / login	Authentication and access to the logged-in area
Dashboard	Entry point for the logged-in user
New project	The project-creation flow
Project result	Displays the generation and the project data
My renovations (history)	Access to previous projects
Properties	Organisation of properties
Rooms	Organisation of rooms
AI Designer	Conversational design experience

Retailer portal	Partner area
Condominiums	Condominium vertical
Layout Studio 3D	Spatial visualisation

Source: prepared by the author (2026).

APIs and functions

The server logic runs in serverless functions grouped by domain. The main groups are summarised below.

Table 24 - API function groups

Group	Main responsibilities	Status
Generation	Image generation, briefing interpretation, computer vision, furniture and mood extraction, anchor-object identification, embeddings and event tracking	Concluded/partial
Agents	Persona chat, comparative panel, debate, synthesis and conversation persistence	Prototype/partial
Image editing	Image edits and anchor/depth-based variations via the fallback provider	Partial
Collaboration	Routes for preference voting and mediation (not exposed in the main navigation)	Prototype
Context	Auxiliary data lookups, such as climate by location	Partial

Source: prepared by the author (2026).

Security and compliance

Because image generation is a costly operation that handles personal data and photos of homes, security and compliance were treated as first-order concerns.

Authentication and access control.

Each user must access only their own data. This is enforced with token-based sessions, Row Level Security (RLS) on every database table, protected routes and a security-definer helper function that checks membership without causing policy recursion. An earlier defect, in which an administrative policy queried its own table and caused infinite recursion (returning an HTTP 500 error on every profile read), was fixed by moving the check into a security-definer function.

Rate limiting and input validation.

Generation endpoints are protected by per-IP rate limiting via a managed Redis service, with an in-memory fallback for local development, and by per-user usage quotas. Input validation guards against malicious prompts, oversized payloads, incomplete data and abusive uploads. For a public document, the specific limits and commercial usage rules are not disclosed.

Data protection (LGPD).

RenovAI handles personal data, images of residential spaces and consumption preferences, which requires attention to the Brazilian General Data Protection Law (LGPD): data minimisation, an explicit purpose, access control, the possibility of deletion and transparency about the use of images.

Table 25 - LGPD measures

Measure	Status
Terms of use published	Concluded
Privacy policy published	Concluded
Data export (portability)	Concluded
Account/data deletion	Concluded
Consent log	Partial/concluded
Encryption of sensitive identifiers	Partial/concluded
Row Level Security on all tables	Concluded
Defined image-retention policy	Planned/to confirm
Notice of non-replacement of a professional	Partial/concluded

Source: prepared by the author (2026).

AI and prompt-injection security.

Because the system uses generative models, there is a risk of prompt injection, jailbreaks and adversarial inputs. Application-level controls are already in production (multi-bucket rate limiting, payload limits, token authentication, RLS, input validation and encrypted identifiers). Recommended hardening that is not yet validated includes separating the system prompt from user input, validating structured output, preventing the model from returning internal instructions, content filtering and a formal penetration test. No formal penetration test has been conducted yet.

External dependencies.

The platform depends on a set of external services, summarised below with their criticality and mitigation.

Table 26 - External dependencies

Service	Role	Criticality	Mitigation
Generative-AI provider	Image generation and text/vision	High	Multi-model fallback to alternative providers
Managed database/auth/storage	Data, authentication and files	High	Standard, portable PostgreSQL; regular review
Serverless hosting	Application execution and deployment	High	Portable functions; alternative providers exist
Managed Redis	Rate limiting	Medium	In-memory fallback in development
Image-editing provider	Variations and edits	Medium	Optional; the main flow does not depend on it

Source: prepared by the author (2026).

Module development

This subsection documents the development of each module: what it does, the steps taken to build it, the main challenges and the current status. The product's maturity is not homogeneous - some modules are in production, others exist as prototypes, internal routes or documented concepts.

Authentication.

The authentication module lets users create an account, log in, access the logged-in area and save their projects; without it the experience would be disposable, with no history or continuity. The development steps were: defining the need for a logged-in area; choosing a managed authentication service; building sign-up and login screens; integrating the user session; protecting private routes; associating projects with the authenticated user; and manual testing of sign-up, login, logout and persistence. The main challenges were balancing early friction (asking for login too soon reduces conversion) with persistence and security (a user must never access another user's data). Status: concluded/partial.

Property, room and version hierarchy.

This module organises projects the way a real home is organised, structuring the work as Property to Room to Project to Versions, so a user can keep, for example, a minimalist, an industrial and a budget version of the same bedroom. The steps were recognising that an isolated 'project' was limiting; defining the conceptual hierarchy; modelling the entities in the database; building screens to list properties, detail a property and show rooms; attaching generations to rooms; introducing the concept of versions; and integrating it with the history. The main challenges were UX complexity (a lay user must not get lost), data migration (legacy projects without a property/room) and making versions easy to recognise and compare. Status: partial/concluded - the concept is consolidated and is one of the foundations of the current version; advanced comparison and versioning are still partial.

Project-creation flow (intelligent intake).

This is the heart of the product: the flow that lets a user start a renovation. The simplified flow is new project, room data, preferences, photo or description, generation, result and history. The steps were building an initial form, defining the minimum briefing

fields, adding room type, desired style and a generic budget band, enabling upload or description of the room, sending the data to the AI layer, showing a processing state, rendering the result, saving it and linking it to the history. The intake collects, among other fields, room type, dimensions, main use, desired style, colour palette, mandatory items, restrictions and a photo. The main challenges were balancing a short form with a complete briefing, avoiding the need for the user to write a technical prompt, handling poor photos, keeping consistency between the briefing and the generated image, and preventing the generation from drifting away from the original room. Status: concluded/partial.

Image generation.

The image-generation module turns the briefing into a photorealistic or conceptual visualisation of the renovated room. It is one of the most important and most difficult components, because a generative model can alter geometry, proportions, doors, windows and elements that should stay fixed. The steps were testing image models, creating initial prompts, separating user data from internal instructions, defining visual styles, adding constraints such as room type and budget, testing perspective and floor-plan outputs, attempting to preserve the room's structure, comparing models, adjusting visual quality and integrating it into the site with storage and display of the result. Output types explored include a photorealistic perspective render (the main deliverable), a generated floor plan (a prototype, with coherence difficulties), before/after (planned), style variations and a visual moodboard. Figure 1 shows a representative result.

Figure 1 - RenovAI result - original room (left) and AI-generated renovation (right)



Source: RenovAI platform (renovai.ia.br), 2026.

The main technical challenges were geometry preservation (the model may move a window, change proportions or invent doors), coherence between the floor plan and the perspective (two individually good images can be mutually inconsistent), scale control (suggested furniture may not fit the real room) and aesthetic quality (making the image look premium and realistic rather than generically 'AI-made'). Because prompt engineering is a differentiator, the specific techniques are kept generic here; at a high level the project used a structured briefing, a separation between mandatory constraints and aesthetic preferences, staged generation, fallback between models, reference images and post-hoc validation of the output. Status: concluded/partial - generation works as the core of the product, but geometric fidelity, scale and automatic plausibility validation are not fully solved.

Furniture identification and recommendation.

This module turns the image or briefing into a list of suggested furniture, ideally compatible with the room, style and budget, so that RenovAI delivers not only an inspirational image but also an understanding of which products could compose the result. The steps were defining the relevant furniture categories, collecting and

structuring an initial catalogue, creating a CSV of products, inserting products into the database, defining standardised fields, testing LLM-based recommendation, experimenting with embeddings, associating products with room types, generating a textual recommendation list and partially integrating it with the project result. Product fields include name, main and sub category, store, current and promotional price, dimensions, predominant material, colour/finish, a short description and a link. The main challenges were price and stock updates, standardisation across stores, missing dimensions, the gap between textual similarity and real compatibility, subjective style classification, linking a real product to the generated image, and scale. Status: partial/prototype - essential to the differentiator but still needing end-to-end validation, automatic catalogue updates, link checking and a stronger connection between the generated image and real products.

Project history.

The history lets the user access previous renovations, review generated images and compare ideas. The steps were defining the need for persistence, associating projects with the user, building a 'my renovations' page, listing projects, showing cards with the main data, opening a project's detail and integrating it with the property/room/version structure. The main challenges were avoiding a confusing history, displaying images without slowness, organising versions and preventing improper access to another user's projects. Status: concluded/partial - a fundamental part of the MVP; future improvements include filters, search and a more visual organisation.

Budget/summary export (PDF).

PDF export was designed to turn a project's result into a shareable document containing a summary of the room, the generated image, the chosen style, the suggested furniture list, notes, a generic estimate, links and next steps. The considered steps were defining the template, selecting the project data, inserting the generated image and the product list, organising generic totals, adding a limitations notice, generating the file, allowing download and testing on mobile and desktop. The main challenges were keeping links up to date, avoiding the appearance of an executive technical document, dealing with changing prices and generating a PDF with heavy images. Status: partial/to confirm - planned and discussed; the final implementation should be confirmed in the repository and described as partial until formally tested.

Architect personas chat.

The personas module simulates different professional perspectives on a room - for example functionality, aesthetics, budget, ergonomics, lighting, circulation and style - so the user sees trade-offs rather than a single answer. The steps were defining the idea of multiple personas, creating agent profiles, defining roles and evaluation criteria, implementing a panel of agents, testing a debate between personas, generating a final synthesis and exploring integration with the AI Designer. The main challenges were excessive complexity and cognitive cost for a lay user, repetition between agents, artificial conflict, and turning the debate into a concrete decision. Status: prototype/partial - the concept and a debate/panel were explored; the main pending item is turning the debate into simple, actionable recommendations.

AI Designer (multi-agent).

The AI Designer is a conversational experience in which the user interacts with agents to build or review a project, allowing a more open conversation than the form-based flow. It can act as a briefing assistant, a style consultant, an alternative generator and a mediator between conflicting preferences. The steps were defining the conversational-designer concept, creating a main agent and a specialist agent, implementing the chat, integrating it with floor-plan or textual input, testing free conversations, creating a panel of agents and attempting to convert the conversation into a structured briefing. The main challenges were maintaining long conversational context, turning conversation into structured data, avoiding generic answers, integrating the chat with image generation and reducing hallucinations. Status: prototype/partial - one of the most promising fronts; the multi-agent version is experimental, while a version with few agents is more viable in the short term.

Retailer Portal.

The Retailer Portal was conceived as an area for partners to register and manage products, connecting RenovAl's catalogue to real furniture. Possible features include store registration, product registration, price and stock updates, image upload, categorisation and integration with recommendations. The considered steps were defining the retailer vertical, modelling the retailer and per-retailer product entities, designing a panel, discussing catalogue import and planning integration with the recommendation flow. The main challenges were the constant change of a store's catalogue, standardisation across retailers, data quality, integration with the generated

room and operational scale. Status: draft/planned - a product front rather than a finished module; an initial interface or documentation may exist, but full operation is not consolidated.

Layout Studio 3D.

Layout Studio 3D aims to represent rooms three-dimensionally, letting the user arrange furniture, view proportions and bring the project closer to a spatial experience. Experiments were carried out with A-Frame and 3D assets, including local tests of multiple rooms and GLB/OBJ models; the production application also uses Three.js with React Three Fiber. The steps were researching web 3D libraries, building a simple 3D scene, inserting walls, floor and furniture, testing ready-made assets, adjusting scale and height, testing navigation and evaluating integration with the AI-generated layout. The main challenges were scale, asset quality and size, performance on phones, the UX of editing in 3D for a lay user, and the difficulty of converting a briefing into a 3D scene. Status: prototype - it validated possibilities but is not fully integrated into the main flow.

Moodboard.

The moodboard is a visual composition of style references, colours, materials, textures, furniture and atmosphere, helping the user understand the aesthetic direction before or after the render. The considered steps were defining its role in the flow, extracting the style and palette from the briefing, generating or selecting visual references, organising blocks of colour, materials and key furniture, connecting it to the final render and allowing comparison between styles. The main challenges were avoiding the misuse of protected images, connecting references to real products, avoiding a generic moodboard and preserving a Brazilian identity. Status: partial/planned - coherent with the proposition; the final implementation should be confirmed and may be treated as a future or experimental feature.

Analytics.

Analytics makes it possible to understand how users use the product, where they drop off and which modules generate the most engagement. For a public document, sensitive acquisition or monetisation figures are avoided; the focus is on the existence and technical planning of events. Useful events include project started, briefing completed, generation started/completed/failed, project saved, version created, product clicked, PDF exported, chat started, room created and property

created. The main challenges were privacy (events must not expose personal content), instrumentation (every screen must record events), interpretation, LGPD transparency and event quality. Status: partial/to confirm - basic events or logs may exist, but a complete metrics-oriented architecture should not be assumed.

Landing page and other fronts.

The institutional landing page communicates the proposition to visitors, explaining the problem, the solution and how it works, and directing the user to start a project; it has gone through continuous visual evolution and may still require copy, branding, responsiveness and performance refinement (status: concluded/partial). Two further fronts were explored but are not finished products: a condominium vertical (Condominium Retrofit), applying the platform to common areas such as halls, gyms and party rooms at a conceptual level and without structural responsibility (status: concept/partial vertical); and preference-mediation ideas for people who decide together, available as a hidden internal route rather than a finished feature (status: concept/prototype). A full marketplace (checkout, cart, payment, order and reputation management) was not implemented and remains a future direction.

Testing and technical evaluation

Testing was predominantly manual and exploratory, appropriate to a lean, fast-moving project, and is documented here honestly - including the testing that was planned but not implemented.

Tests performed.

Manual flow tests were run throughout development to validate that screens and flows worked (sign-up/login, project creation, form completion, sending data to the AI, image return, result display, saving, navigation to the history and opening a previous project). Image generation was tested with real or semi-real cases across different rooms, styles and requests, which surfaced problems such as spatial inconsistency, undue changes to the structure, excessive decoration, incompatible furniture and visually good but impractical results. Catalogue experiments (CSV, database seed and recommendation) revealed the need to standardise categories, inconsistent store data and the need to validate embeddings and semantic search. The 3D/AR experiments with A-Frame revealed scale, performance, asset-quality and integration difficulties. Exploratory validation with users and potential users was also carried out; any

percentages or samples used in this work are taken directly from the original research and cited accordingly, and no metrics are invented here. The platform has generated 67 projects in production with completed generation, validating the end-to-end pipeline (briefing interpretation, image generation, furniture extraction and storage).

Automated test suite.

There is no robust automated suite of unit, integration and end-to-end tests covering the whole application. For honest documentation, the status of each test type is summarised below.

Table 27 - Status of automated testing

Test type	Status
Unit tests	Not implemented / to confirm
Integration tests	Not implemented / to confirm
End-to-end tests	Not implemented / to confirm
Automated API tests	Not implemented / to confirm
Automated image-generation tests	Not implemented
Visual-regression tests	Not implemented
Automated accessibility tests	Not implemented
Automated security tests	Not implemented
CI/CD with a full suite	Not implemented / to confirm

Source: prepared by the author (2026).

Tests planned but not implemented.

Several types of testing were considered but not implemented, and are listed here as part of an honest account of the project's maturity and as a basis for future work.

Table 28 - Planned but not implemented testing

Test	What it would cover	Status
Unit tests	Isolated functions: form validation, briefing normalisation, AI payload building, product formatting, permissions	Planned/not implemented
Integration tests	Frontend-backend, backend-database, backend-AI, auth and private routes, generation and storage	Planned/not implemented
End-to-end tests	Full flows: sign-up to project to generation to history; login to property to room to version; project to generation to PDF	Planned/not implemented
Load and stress tests	Many simultaneous generations, response time, external-API failures, behaviour at usage limits	Planned/not implemented
Security penetration test	Improper project access, auth flaws, malicious upload, data exposure, prompt injection, key leakage	Planned/not implemented
Accessibility tests	Contrast, keyboard navigation, form labels, alt text, screen readers, error states	Planned/not implemented
Cross-device/browser tests	Chrome desktop, Safari iOS, Chrome Android, Edge, Firefox, different screen sizes, mobile upload	Partial/not formalised
Closed beta	Comprehension of the proposition, image quality, usefulness of recommendations, sign-up friction, perceived trust	Planned/not implemented
A/B tests	Short vs complete form, photo before vs after briefing, landing CTA, chat vs form, one vs multiple versions	Planned/not implemented

Source: prepared by the author (2026).

QA process.

Quality assurance was mostly manual and iterative, following the cycle below.

QA CYCLE

Implementation

- > manual test
- > bug identification
- > fix (with AI/code)
- > new manual test
- > visual validation
- > deployment
- > new test in production

Table 29 - Observable bug types

Type	Example
Visual	Broken layout on mobile
State	A project does not appear after generation
AI	The image ignores the briefing
Database	An incorrect project/user relation
Upload	A photo does not load
Performance	A screen is slow with no feedback
Deployment	A build fails due to a dependency/configuration
Route	A page is accessible directly without context
UX	The user does not understand the next step

Source: prepared by the author (2026).

2.5 The Business Plan

The business plan consolidates the market, business model, go-to-market and financial feasibility of RenovAI at a strategic level.

Market and Competitor Analysis (SWOT).

Table 30 - SWOT matrix

Strengths	Weaknesses
Native localisation for the Brazilian market (language, prices, brands); mobile-first experience; low operating cost; product already in production.	Early-stage product; revenue model still under validation; lean team; dependence on third-party AI services.

Source: prepared by the author (2026).

Table 31 - SWOT matrix (continued)

Opportunities	Threats
Public housing-renovation credit programmes; fast-growing AI design segment; falling cost of generative AI; a large category validated internationally yet under-served in Brazil; non-dilutive public funding.	Entry of better-funded competitors; commoditisation of generative AI; exchange-rate pressure on cloud costs; restrictions in some retail affiliate programmes.

Source: prepared by the author (2026).

Business Model (Business Model Canvas).

Table 32 - Business Model Canvas

Block	Content
Customer Segments	General-public renovators across income levels; retailers and developers (B2B).
Value Propositions	See your renovation before buying: photorealistic visualisation + furniture list, free, in minutes, no technical knowledge.
Channels	Mobile-first web (renovai.ia.br); organic SEO and social media; referral; retail partnerships.
Customer Relationships	Self-service product; conversational assistant; e-mail support; native sharing.
Revenue Streams	Commissions on referred sales; optional paid plans; future marketplace.
Key Resources	Generative-AI pipeline; product and engineering; user base with saved projects.
Key Activities	Product development; AI integration; partnership onboarding; curation.
Key Partnerships	Cloud and AI providers; furniture retailers; potential public funding agencies.
Cost Structure	Variable cloud-AI usage and a lean fixed-cost base; no inventory or logistics.

Source: prepared by the author (2026).

Marketing and Sales Strategy.

The go-to-market is fully digital, with no human sales force. Acquisition combines organic channels (search, social media and referral) with retail and developer partnerships. Paid media is introduced only once the commission revenue can sustain the cost of acquisition.

Financial Projection and Feasibility.

The model combines a low, cloud-native fixed-cost base with commission-based revenue and optional subscriptions. The variable cost per use is modest and has been falling with the broader decline in generative-AI costs. Feasibility depends on validating user conversion with proprietary data; the lean cost base provides runway to do so. The initial investment was bootstrapped by the founder, with non-dilutive public grants under consideration.

2.6 Validation and Results

Validation followed a lean, evidence-based approach, using the product in production with a controlled user base to test the business hypotheses rather than to claim market traction.

Validation Methodology.

The MVP was released in production and instrumented with funnel analytics (sign-up, wizard completion, successful generation, sharing, PDF export) under explicit LGPD consent, with UTM attribution by channel.

Market Validation Results.

As of May 2026 the platform had 20 registered users and 8 users who generated at least one project, totalling 67 completed generations. The activation rate (sign-up to first generation) was 40%, identifying the activation funnel as the main bottleneck. These figures constitute deliberate technical validation, not statistical market traction.

Pivoting or Persisting.

Validation refined the revenue assumptions and led to the introduction of optional paid plans alongside the commission model, while preserving the core thesis rather than abandoning it.

Table 33 - Key performance indicators monitored

KPI	What it measures	How to measure
Registered users	Reach	Database
Active users	Real product usage	Database
Activation rate	Sign-up to first generation	Funnel analytics
Click-through on products	Purchase intent	Link tracking
Conversion to paid plan	Monetisation	Subscription data

Source: prepared by the author (2026).

Risks and Mitigation Plan.

- **Financial:** the revenue model requires validation with proprietary data; mitigated by a lean cost base and optional paid plans.
- **Technological:** dependence on third-party AI services; mitigated by provider redundancy in the pipeline.
- **Competitive:** a well-funded incumbent could enter; mitigated by localisation and product focus.
- **Legal:** data protection (LGPD) and tax treatment of revenue; mitigated by basic compliance already implemented and planned legal review.

Table 34 - Technical risk analysis

Risk	Probability	Impact	Mitigation
AI provider price or policy change	Medium	High	Multi-model architecture with fallback
External API unavailability	Medium	High	Provider redundancy and error handling
Geometric inconsistency in generation	High	Medium	Structured briefing, staged generation, future validation
Outdated catalogue/links	High	Medium	Planned automatic updates and link checking
Lack of automated tests	High	Medium	Planned test suite; manual QA in the interim

Security / prompt injection	Medium	High	Application controls in production; planned penetration test
Exchange-rate impact on AI cost	High	Medium	Structural fall in AI cost; provider flexibility

Source: prepared by the author (2026).

Technical challenges and lessons learned

The development surfaced a set of technical challenges whose resolution shaped the product. They are summarised below, each with the main lesson learned.

Challenge 1 - A beautiful generation is not a useful generation.

Visual quality alone is not enough: an image can look sophisticated and still be poor for planning if it changes the structure of the room, suggests furniture that does not fit, ignores constraints, invents elements or exceeds the budget. The lesson is that the AI must be evaluated by usefulness, not only by aesthetics.

Challenge 2 - Geometric fidelity.

Preserving the geometry of the room is difficult: most generative models tend to embellish a space but may alter its structure. Mitigations considered include a more structured briefing, collecting measurements, using a reference image, staged control and post-hoc validation. Resolution status: partial.

Challenge 3 - Floor plan and render do not always agree.

Generating a coherent floor plan and perspective is harder than generating a single image, because the system must understand the space consistently. The lesson is that a floor plan requires a more structured approach, possibly with an intermediate geometric model, rather than purely visual generation.

Challenge 4 - Recommending real furniture.

Recommending real products requires far more than listing attractive items: it must consider measurements, style, price, availability, store, material, colour, lead time, compatibility with the room and substitutes. The lesson is that the catalogue must be treated as a data system, not merely as text.

Challenge 5 - A lay user cannot write a prompt.

RenovAI cannot depend on users writing detailed commands; the product must extract information through the interface, questions and guided choices. The lesson is that the intake is as important as the AI model itself.

Challenge 6 - Multi-agent can help or hinder.

Agents and personas enrich the analysis but can also make the experience complex; a lay user may not want to follow a debate between ten specialists. The lesson is that a multi-agent approach should be used internally or in summary form, delivering an actionable synthesis.

Challenge 7 - Technical and legal scope.

RenovAl must avoid any promise of a structural technical project, which directly influenced the product's positioning. The lesson is that explicit limits increase safety and credibility.

Challenge 8 - The product must not become a collection of demos.

Many fronts emerged during the project (3D, AR, retailer, condominiums, personas, chat, catalogue, marketplace); the challenge is to keep focus. The lesson is that the product's core must remain clear: helping the user visualise, decide and plan a light renovation.

Challenge 9 - Dependence on external APIs.

AI services change price, availability, latency, quality and policy, which makes the architecture vulnerable if it relies on a single provider. The lesson is that the application must abstract AI providers and allow fallback, without exposing sensitive detail.

Challenge 10 - Documentation in a solo project.

Because the project evolved quickly, not every decision was documented when it was made. The lesson, for this work, is the need to reconstruct the trajectory honestly, marking status and uncertainties.

Features and ideas not implemented

An honest account of the project includes the features that were ideated, prototyped or abandoned. They are documented here both for transparency and as a basis for future work; the table separates what was done from what was not.

Table 35 - Features and ideas not (fully) implemented

Idea	What was done	What was not done	Status
AR/VR	Technical experiments with A-Frame, local 3D scenes and assets	Full integration into the main flow, stable mobile AR, a real 3D catalogue	Technical prototype

Condominium Retrofit	Vertical ideation, a possible dedicated section, use-case discussion	A complete system for managers, a collective-approval flow, technical budgeting	Concept/partial
Iterative AI Designer	Conversational chat and agents explored	A robust image-editing loop, consistent geometry across iterations, a visual diff between versions	Partial/prototype
Persona voices	Textual personas and debate explored	Real audio, synthetic voices, a multimodal experience	Idea/not implemented
Full marketplace	Initial catalogue, product recommendation, retailer-portal idea	Checkout, cart, payment, order and reputation management, catalogue reconciliation at scale	Not implemented
End-to-end semantic search	Embedding experiments, an initial product base, recommendation tests	Systematic relevance validation, benchmarking against filters, automatic base updates	Partial/not validated
Robust anti-prompt-injection	Consideration of validations, conceptual separation of internal instructions and user input	A formal penetration test, an adversarial-attack suite, complete server-side validation, an external audit	Partial/not validated
Native mobile app	Mobile-first web product	An iOS/Android native application	Planned/not implemented

Source: prepared by the author (2026).

3 Conclusion

This work developed and validated RenovAl, a generative-AI platform that lets any Brazilian visualise a renovated room and obtain a furniture list for free, and elaborated the corresponding business plan. The objectives were met: a functional MVP is in production, validated technically with 67 generated projects; a revenue model was defined; and a complete business plan, including market sizing, competitive analysis and financial feasibility, was produced.

The central finding is that the solution is technically robust and the business model is feasible, contingent on validating user conversion with proprietary data. The platform's lean, cloud-native cost base provides runway for that validation.

Future work includes formalising the company, validating the revenue assumptions with real data, expanding the product across platforms, and growing the user base. The technical roadmap below complements these steps.

Future technical work

Beyond the business steps, a clear technical roadmap emerged from the development. The priorities below are described without business strategy.

Consolidate the current architecture.

Before expanding new features, it is advisable to consolidate the real folder structure, the database entities, the relationship between projects, rooms and versions, the AI layer, logging, permissions, error handling and deployment. Priority: high.

Implement an automated test suite.

The platform should evolve to include automated tests, prioritised as follows: unit tests of critical functions (high), integration tests with the database (high), end-to-end tests of the main flow (high), security/penetration testing (high), visual-regression and accessibility tests (medium) and load/stress tests (medium).

Improve geometric fidelity.

Possible technical paths include mandatory collection of measurements, the use of a simplified floor plan, detection of structural elements, segmentation of the room, layout control via JSON, automatic object validation, comparison between the original and generated image, and staged generation.

Create an intermediate layout model.

Instead of generating an image directly from text, the system could first produce a structured layout that acts as a bridge between intent, image, products and 3D, for example:

```
{
  "room_type": "bedroom",
  "dimensions": { "width": "to_confirm", "length": "to_confirm" },
  "fixed_elements": ["door", "window"],
  "furniture": [
    { "type": "bed", "position": "left_wall", "size": "double" }
  ]
}
```

Evolve product recommendation.

Next steps include normalising the catalogue, validating links, updating availability, classifying products by style, considering dimensions, improving room matching, combining traditional filters with semantic search and explaining each recommendation.

Other technical priorities.

Further work includes implementing version comparison (side by side, with the main differences and a generic cost level); strengthening the AI Designer (a conversational briefing, project memory, adaptive questions and persona synthesis);

improving observability (generation and error logs, response time, a technical-failure rate and exception tracking); maturing security and LGPD (reviewing the privacy policy and terms, image-retention limits, reinforced RLS, upload validation and prompt-injection testing); improving accessibility (input labels, contrast, keyboard navigation, focus states, alt text and screen-reader support); improving performance (image compression, lazy and skeleton loading, result caching and timeout handling); integrating 3D progressively (from a structured 2D layout to a simple visualisation and an optional AR mode); and producing technical documentation (architecture, data model, generation flow, an internal prompt guide without exposing differentiators, a security guide, a test guide, a module map, discarded decisions and a deployment manual).

Limitations of the work

This work has limitations that should be made explicit. The validation was deliberately technical, with a controlled and small user base, so the conversion and market hypotheses still require broader testing; the automated-test coverage is minimal, which increases the risk of regressions; several modules remain at the prototype or concept stage, with non-homogeneous maturity; the architecture evolved over time, which makes some older decisions harder to trace; and the platform depends on third-party AI services whose price, availability and behaviour may change. These limitations do not invalidate the contribution of the work - a functional MVP, a documented architecture and process, and an honest map of what was built, planned and discarded - but they delimit the scope of the conclusions and orient the future work described above.

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Appendices

This appendix consolidates supporting material for the work: a summary of the modules and their status, a glossary of terms and a list of items to be confirmed against the repository.

Appendix A - Consolidated module and status summary

The table below consolidates the modules and fronts of RenovAI with their current maturity.

Table 36 - Modules and status

Module	Status	Note
Landing page	Concluded/partial	Public site live and evolving
Authentication	Concluded/partial	Required for the logged-in area
New project	Concluded/partial	Main MVP flow
Project result	Concluded/partial	Displays the generation and data
History	Concluded/partial	Allows returning to projects
Property	Partial	Higher-level organisation
Room	Partial	Organisation by room
Versions	Partial	Fundamental for comparison
Image generation	Concluded/partial	Core of the product
Floor plan	Prototype/partial	Coherence still a challenge
Furniture recommendation	Partial/prototype	Needs validation
Catalogue	Partial/prototype	Base structured, maintenance pending
Budget PDF	Partial/to confirm	Verify final implementation
AI Designer	Prototype/partial	Conversational

Personas	Prototype	Debate/evaluation
Retailer Portal	Draft/planned	Not consolidated
Condominiums	Concept/partial	Explored vertical
Layout Studio 3D	Prototype	Technical experiments
AR/VR	Prototype	Not a final product
Moodboard	Partial/planned	Verify implementation
Analytics	Partial/to confirm	Robustness not confirmed
Marketplace	Not implemented	Concept/future direction
Native app	Not implemented	Possible future

Source: prepared by the author (2026).

Appendix B - Glossary of terms and acronyms

Table 37 - Glossary

Term	Meaning
AI	Artificial intelligence
LLM	Large language model
MVP	Minimum viable product
RLS	Row Level Security (per-row database access control)
LGPD	Brazilian General Data Protection Law
API	Application Programming Interface
CDN	Content Delivery Network
TAM/SAM/SOM	Total / Serviceable Available / Serviceable Obtainable Market
SWOT	Strengths, Weaknesses, Opportunities and Threats
BMC	Business Model Canvas
UX	User experience
CSV	Comma-separated values (a tabular data format)
Embedding	A numerical vector representing the meaning of a text or image
Prompt injection	An attack that manipulates a model through crafted input
Fallback	An automatic switch to an alternative provider on failure
GLB / OBJ	3D model file formats
Briefing	The structured set of the user's preferences and constraints

Source: prepared by the author (2026).

Appendix C - Items to be confirmed in the repository

Before turning this report into the final chapter, the following items should be confirmed against the current repository:

- the exact current stack;
- the real folder structure;
- the technologies effectively in production;
- the real database tables;
- the real route names;
- the status of PDF export;
- the real status of the moodboard;
- the real status of analytics;

- the automated-test coverage, if any;
- the RLS policies;
- the existence of terms of use and a privacy policy;
- logging and observability;
- the active AI integrations;
- which modules are public, private or accessible only by a direct URL.

Annexes

No annexes (third-party documents) are included in this version.